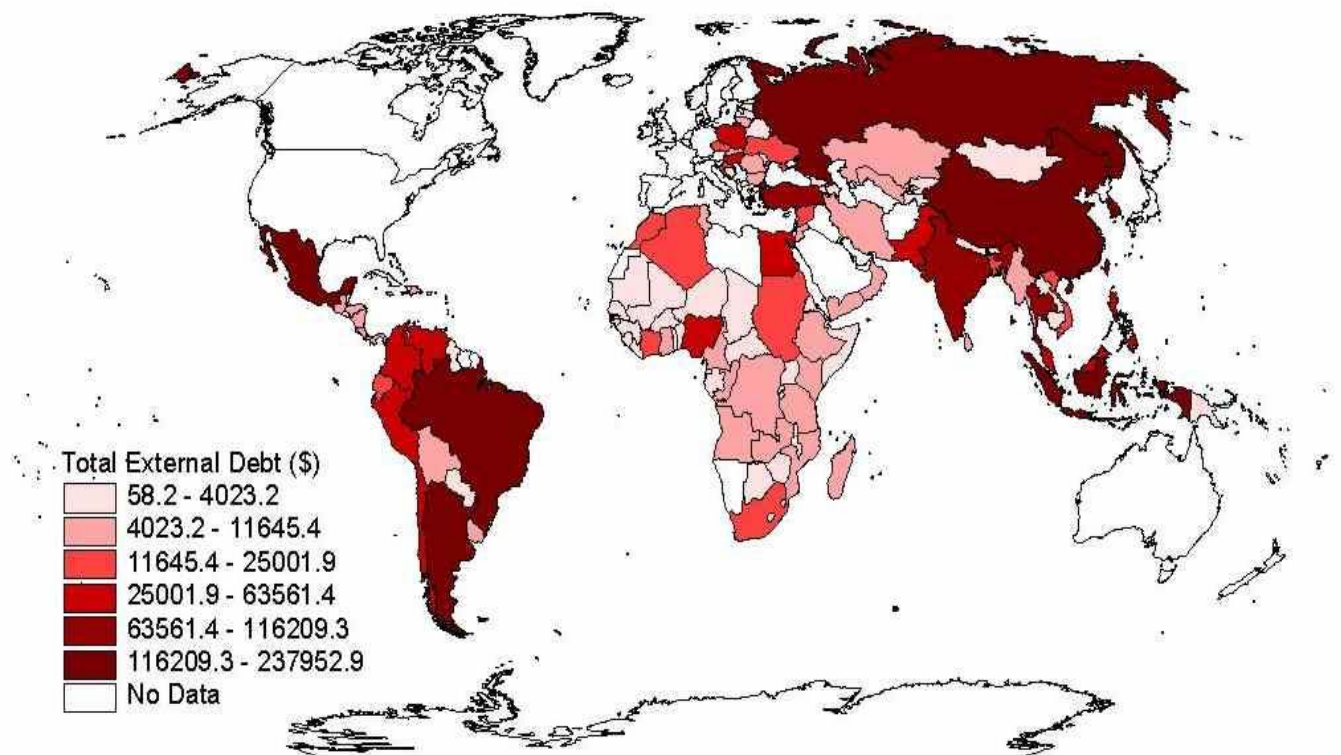




# Global Economy

## Total External Debt



## Arc View Lesson

## GIS Information

### Here's a list and explanation of important features in ArcView:

**Project :** In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

**View**  : A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

**Table of Contents:** All themes in a View are listed to the left of the map. The Table of Contents shows the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

**Theme:** A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

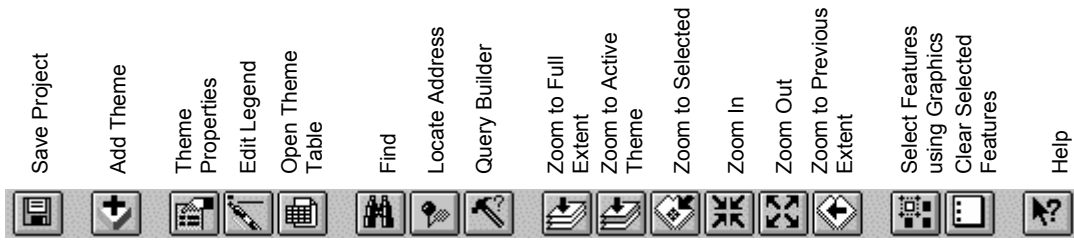
**Tables**  : Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

**Layout**  : The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

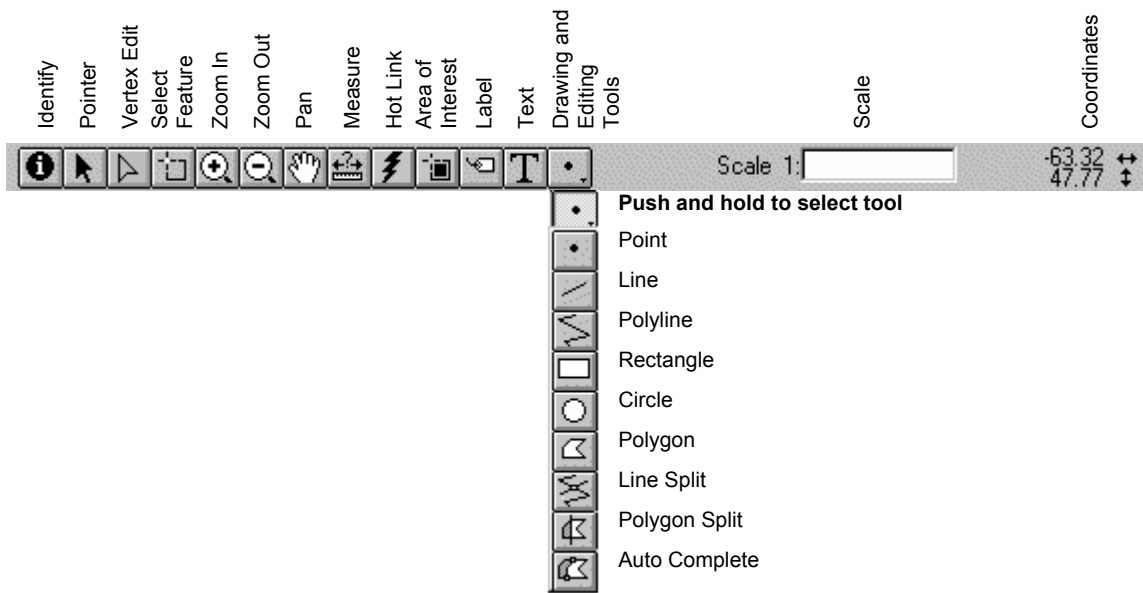
**Hotlink:** In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

## ArcView Buttons Reference Guide

### View - Buttons



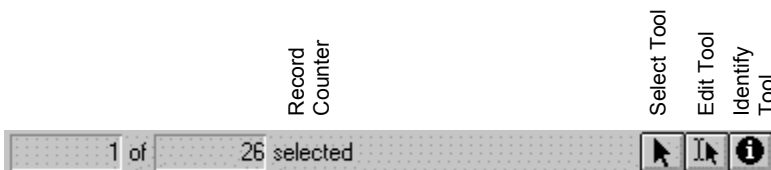
### View – Tools



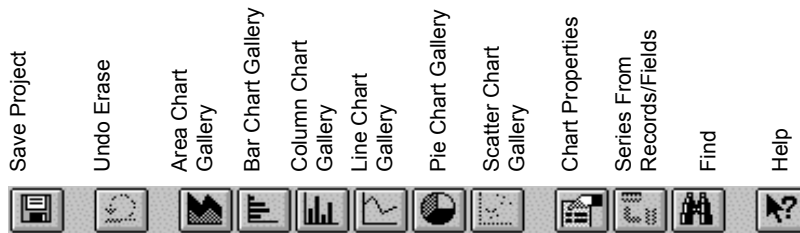
### Table - Buttons



### Table - Tools



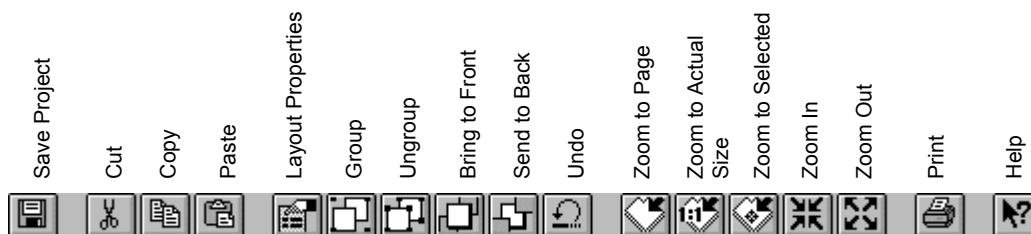
## Chart - Buttons



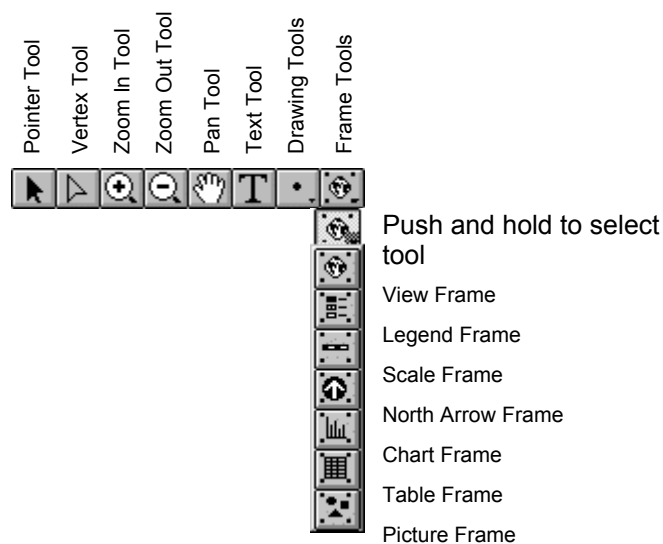
## Chart - Tools



## Layout - Buttons



## Layout - Tools



For more Information: <http://www.esricanada.com>

## Getting Started

**Before you begin**, the lesson data folder must be **copied** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

**!** (The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. The data are downloaded as a self-extracting file. Double click the file in windows explorer to extract all the data files. Ask your teacher where to find the data folder that has been downloaded).

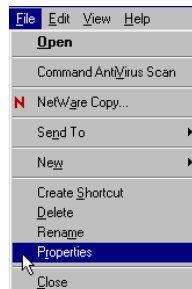
To do this, use the file manager program on your computer. For Windows-based systems this will be **My Computer / Windows Explorer /**.

Once the data folder has been **copied** to the required destination the files must be made usable by removing the **read only** attribute. For Windows-based systems follow these instructions:

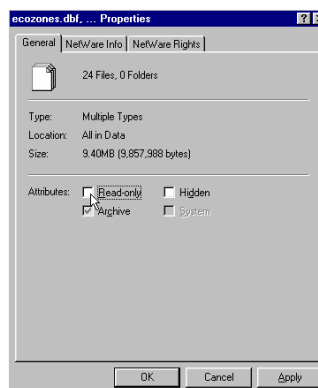
**!** (The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)

1. Open your new data folder.
2. Choose **Edit > Select all** (they will all appear Blue)

3. From the menu choose:  
**File > Properties**



4. Remove the check mark in the box beside the label **read only**.

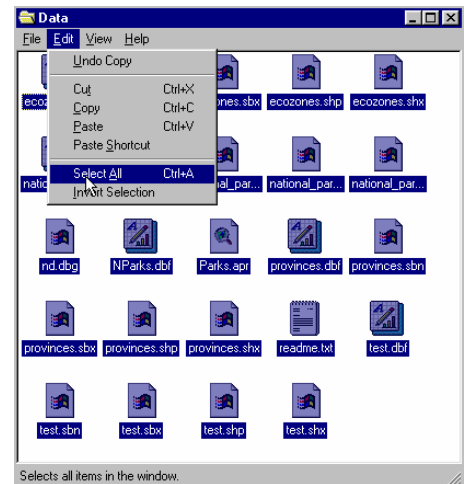


5. Click **Apply**, and then **OK**.

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

**!** Denotes an observation.

**?** Denotes a question to be answered on the Student Activity Sheet.



## STARTING THE PROJECT

This lesson will emphasize the use of Geomatic data to study several aspects of the global economy, gross domestic product, income disparity, poverty, external debt, and foreign investment.

### STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **O.K.**

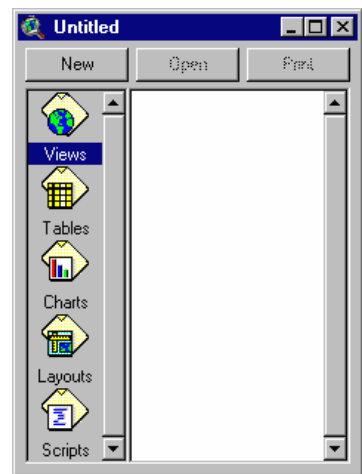


When the program starts, it automatically opens a project window titled **"Untitled."**

If you like, you can maximize the **ArcView** window to fill the screen.

**!** In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.

3. Now you are ready to add documents to your new project.
4. With the **"Views"** icon selected, click on **New** and maximize the **View1** window.

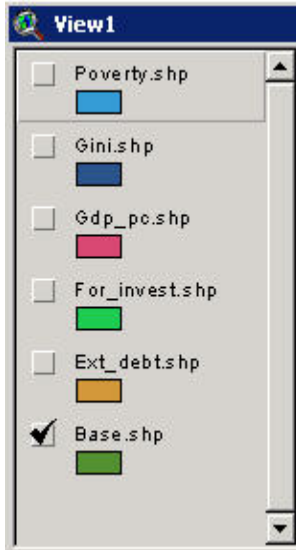


5. Click the **Add Theme**  button and navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

*base.shp*  
*ext\_debt.shp*  
*for\_invest.shp*  
*gdp\_pc.shp*  
*gini.shp*  
*poverty.shp*

6. Click **O.K.**

This will add these six themes to the table of contents of the new view. They are now available for display and editing.



7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. Click on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by clicking on other themes.

9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project**. Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **.apr** means that you are saving your work as a project that can be opened up in the future).

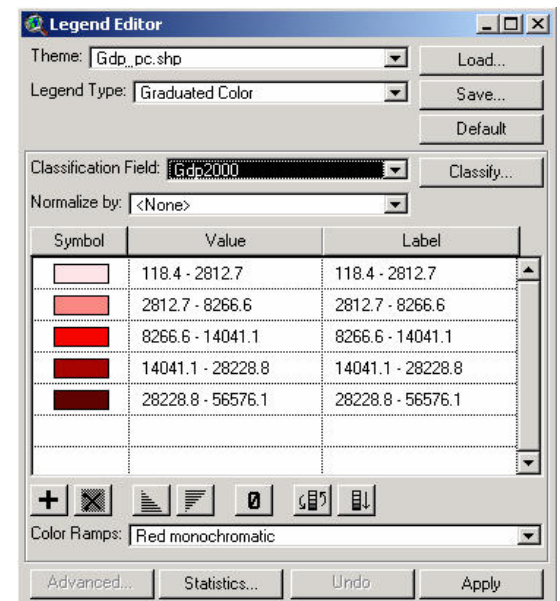
## ArcView - DISPLAYING A MAP

### Mapping Gross Domestic Product Per capita

1. From the **Project window** select the **Views** icon.
2. Select and open **View1**.
3. **Double Click** the **gdp\_pc.shp** theme to edit it.
4. Open **Legend Type** list.
5. Select **Graduated Colour**.
6. Open **Classification Field** list.
7. Select **gdp2000**
8. Click **Apply**.
9. Close **Legend Editor** window.

! This will display the gross national product per capita for each country for 2000 in Constant 1995 US dollars per person.

Data are missing for some countries.



? Can you see and describe the distribution of developed countries, developing countries and under developed countries?

Use the zoom button  to look at Europe.

? Describe the pattern in the GDP you see in the European region?

Use the Identify Tool to  query a country to get its name and other details.

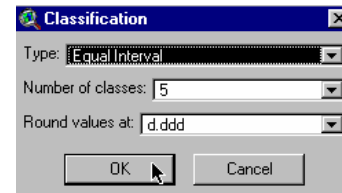
Redisplay the map using different years. E.g. (gdp1995 or gdp1987)

? How do the maps change? What are some possible reasons for the changes?

## ArcView - CLASSIFYING A MAP

1. Double Click the *gdp\_pc* theme.
2. Click **Classify**.
3. Select **Type Equal Intervals**.
4. Click **O.K.**
5. Click **Apply**.
6. Close **Legend Editor** window.

? How did the map and legend change?



Repeat this process using different classification types.

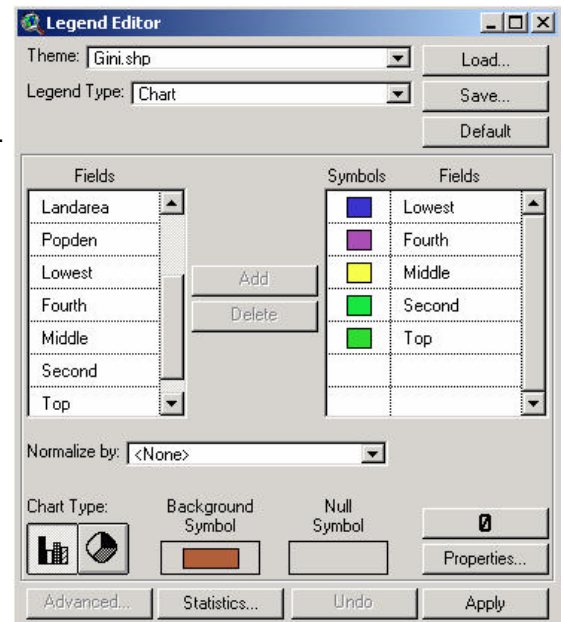
? How did the map and legend change with the different classification types?

? Why is it important to select the proper classification?

## ArcView - DISPLAY DATA AS CHARTS

We will now display income disparity using charts.  
The data used show the share of the total income for the lowest, fourth, middle, second and top 20% of the population.

1. Double Click *Gini* theme.
2. Open **Legend Type** list.
3. Select **Chart**.
4. In the **Fields List**, Select **Lowest**.
5. Click **Add**.
6. Repeat for **Fourth**, **Middle**, **Second**, and **Top**.
7. For **Chart Type** Click **Bar Chart**.
8. Double Click **Background Symbol** and Select a **Background Symbol** that does not conflict with your legend colours.

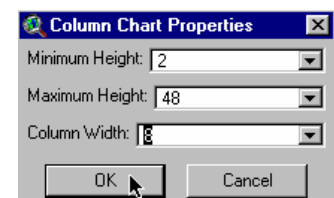


9. In the **Legend Editor** window, Click **Properties** to alter the look of the charts.

10. Click **OK**.

11. Click **Apply**.

12. Close **Legend Editor** window.



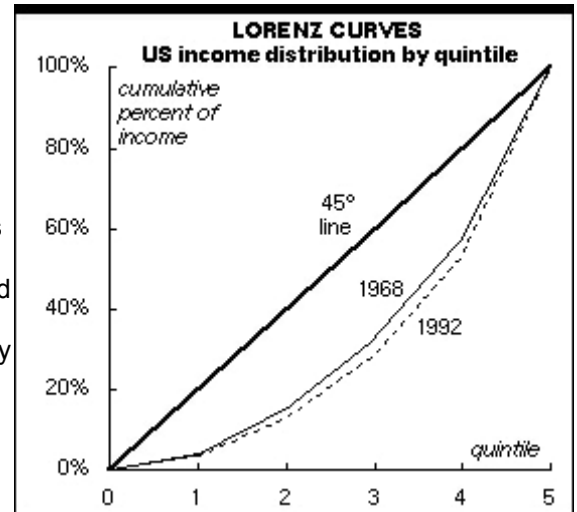


? By zooming in and comparing the charts, can you describe the graphs of several countries?

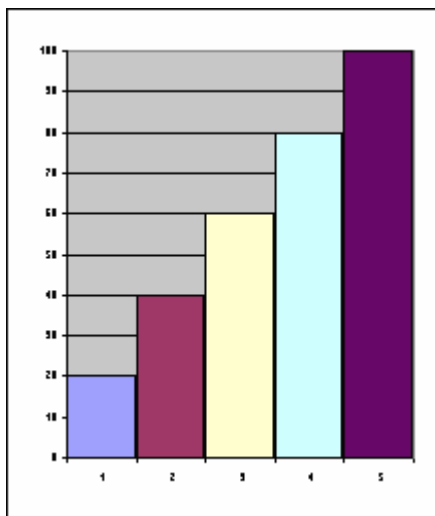
? What does this tell you about income equity?

! Plotting distribution: Lorenz curves and GINI index:  
For a more detailed explanation of the GINI index, please check the book "Global Connections: Canadian and World issues, Bruce Clark and John Wallace, Prentice Hall, 2003 pages 203-204.

We can start with the nearby graph, derived from the numbers in the table. The horizontal axis shows the quintiles of population; the vertical, the cumulative share of income earned by the plotted quintile and the ones below it. For example, in 1968, the cumulative value for quintile 1 is the share earned by quintile 1, or 4.2%; that for quintile 2, 11.1%+4.2%, or 15.3%; and so on, up to quintile 5, when the sum is 100%. The graph plots these values for 1968 and 1992. These lines are called Lorenz curves.



In a society with perfectly equal income distribution, the cumulative share of income would be equal to the cumulative population share: e.g., the value for quintile 2 would be 40% (20%+20%), instead of 1968's actual 15.3%. This mythical egalitarian society is represented by the heavy 45° line. A bar chart similar to those on the map for the mythical egalitarian society would look like the sample chart below.



The gap between the actual lines and the mythical line is a function of the degree of inequality; the 1992 line is further away from the heavy line than 1968's is. The Gini index measures the gap between the actual line and the 45° line. In the egalitarian society, the Gini would be 0.000, since the Lorenz curve would match the 45° line perfectly; the higher the Gini, then, the greater the distance, and the more unequal the distribution of income. In a perfectly unequal society, in which one person (or household or family) had all the income, the Lorenz curve would look like a backwards "L," and its value would be 1.000. In practice, the Gini usually falls between 0.200 and 0.450. The U.N. uses a Gini value of 0.4 as a warning level that countries of economic disparities are becoming excessive.

? Select several countries and compare the charts and data with the mythical ideal graph presented above. What can you say about the income inequity in each of the selected countries?

Use the Identify Tool to  query five countries to get the data needed to prepare Lorenz curves.

? After plotting the data as Lorenz curves can you describe the income equity for each of the countries?

## ArcView - BUILDING A QUERY

We will now build a query to explore the level of external debt held by each country. The data are presented (in Million current US dollars)

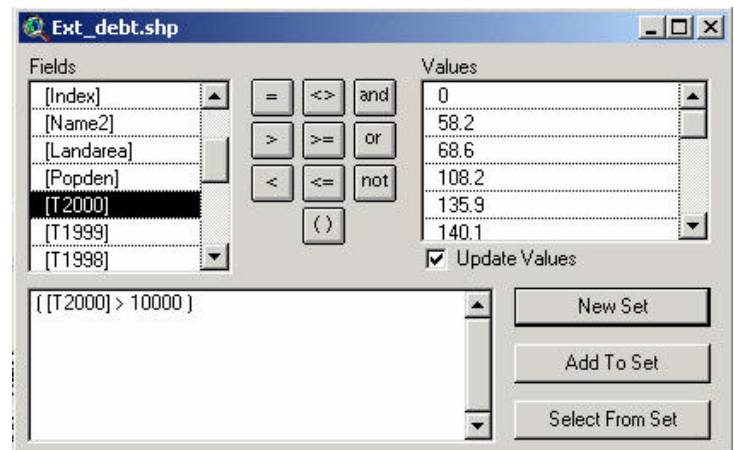
1. Double Click the **ext\_debt.shp** theme to edit it.
2. Open **Legend Type** list.
3. Select **Graduated color**.
4. Open **Classification Field** list.
5. Select **T2000**.
8. Click **Apply**.
9. Close **Legend Editor** window.

! This layer has the total external debt carried by each country 1987 to 2000.

Now let's build a query to find all the countries with a total external debt greater than \$10 billion in 2000.

1. Select and activate **ext\_debt** theme.

2. Click **Query Builder** button. 
3. Double Click Field **T2000**.
4. Click the > button.
5. Enter **10000** in the equation.
6. Click **New Set** button.
7. Close **query** window.



? List five countries that had a total external debt greater than \$10 billion?

? Describe the global pattern of external debt over \$10 Billion in 2000?

8. Click **Clear Selected Features** button. 

Using the same process as used for 2000, choose other years and or debt limits.

9. Click **Query Builder** button. 
10. Double Click Field **T1987**.
11. Click the > button.
12. Enter **15000** in the equation.
13. Click **New Set** button.
14. Close **query** window.

? List five countries that had a total external debt greater than \$15 billion in 1987?

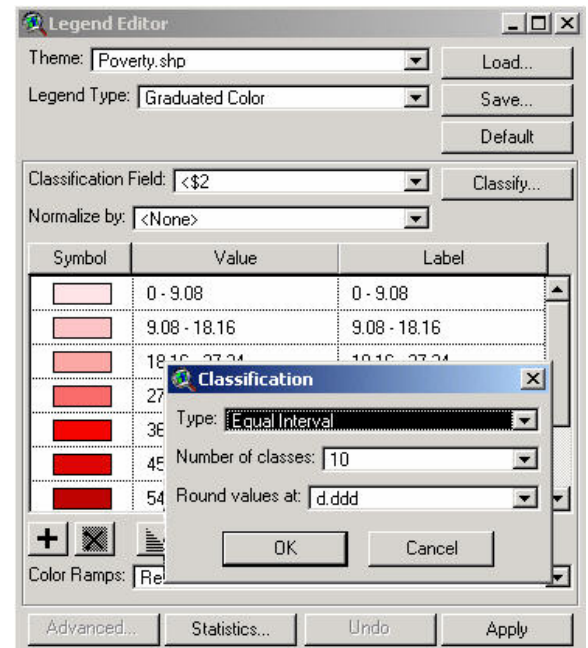
? Describe the global pattern of external debt over \$15 Billion in 1987?

Let's build more queries to explore total external debt.

## ArcView - DISPLAY PERCENTAGE DISTRIBUTION OF POVERTY

We will display poverty as the percentage distribution of country population earning < \$2 and < \$1 a day 1994 to 1999.

1. Double Click **Poverty** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select **<\$2**.
6. Click **classify** button.
7. Open **Type** list.
8. Select **Equal Intervals** and **10 classes**.
9. Click **OK**.
10. Click **Apply**.
11. Close **Legend Editor** window.



? What can you say about the global distribution of Population living below \$2/day?

? Are there regions not represented?

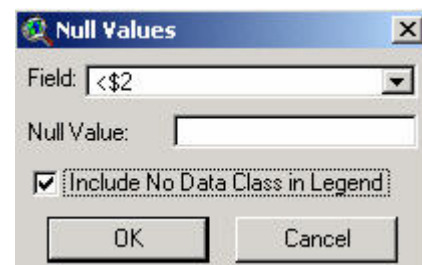
? Why?

! To have a completed map, we will put a no data item in our legend for those countries that do not have representation.

12. Double Click **poverty** theme.

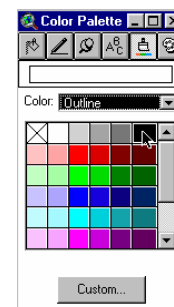
14. In the **Legend Editor** window, Click on the **Null Value** button.

14. Check **Include No Data Class in Legend** box
15. Click **OK**.



16. Double Click no data legend item.

17. Click **The Colour Palette** button.
18. Select a new colour for the **no data** legend item.
19. Open **The Colour** list, **Select Outline**.



20. Select **black** for outline colour.
21. Close **Colour Palette** window.
22. Click **Apply**.
23. Close **Legend Editor** window.

**!** Using the same process that was used to display Population living below \$2/day, display the Population living on less than \$1/day. Explain the distribution patterns you see.

**?** Explain the distribution patterns you see for poverty.

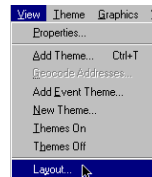
**?** How does it compare with the map for \$2/day?

## ArcView - MAP LAYOUT FOR PRINTING

Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

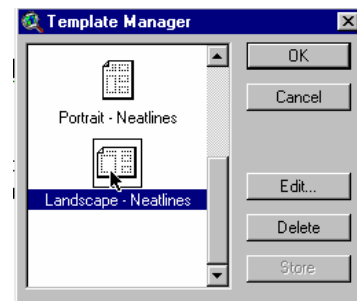
### SET UP MAP

1. Click Menu **View**.
2. Click **Layout**.



3. Select the **Layout Template** you want to use (One of the landscape templates is suggested) .

4. Click **O.K.**
5. Maximize the Layout screen.



### EDIT TITLE

1. Double click on the title **View1**.
2. Edit the title to reflect the purpose of the map, e.g., poverty, and GDP.
3. Click **O.K.**
4. Resize the title using the corner handles and move it into a position that is visually balanced.
5. Click and drag to reposition the legend if necessary.

### EDIT NORTH ARROW

1. Double click on the *north arrow* and select a different presentation. Ensure that the arrow is pointing in the representative north direction of the map.
2. Click and drag to reposition or rotate (using rotation function) as needed.

### EDIT SCALE BAR

1. Double click the *scale bar*.
2. Choose a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. Change the units to **kms**.
4. Find a combination of style, scale and divisions that is visually pleasing.

5. Click and drag to position to an appropriate location.

### ***ADD LABELS, TEXT, OTHER INFORMATION***

1. Click on the **Text** tool.
2. Click at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., "produced by (your name, date)".
4. Select desired formatting information such as **alignment**, **spacing** and **rotation**.
5. Click **O.K.** when it appears, as you want it.
6. Click on **Pointer** tool.
7. Select the new text
8. Click once to change location or size. Click twice to change content.

**!** Experiment with all of the components of the layout until you have the look you want.

### ***CHANGE THE NAME OF THE LAYOUT***

You only have to do this if you want to save your project on the computer or on a disk.

1. Click menu **Layout**.
2. Select **Properties**.
3. Change the "name". Click **O.K.**

### ***ARRANGING YOUR LAYOUT***

**!** Experiment with the layout buttons (use the tools page provided at the beginning of this lesson if you are unsure of their function).

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** ( change order that pieces were laid down, e.g., lines on top of polygons)
- **send to back** ( change order that pieces were laid down, e.g., polygons behind lines)
- **zoom** ( make image bigger or smaller)
- **neat line** (border around layout)

### ***PRINT MAP LAYOUT***

1. Click **Print** button.
2. Click **Setup**.
3. Select your printer properties as outlined by your teacher.
4. Click **O.K.** to print.

**!** **Note** that the map can also be saved to a printer file.